

Data Analytics (DA)

Module-End -2

Module End Assignment:

“Data Transformation, Modeling & Visualization with Power BI”

Title: Analyzing Smart City Bike Sharing Data using Power BI

Problem Statement:

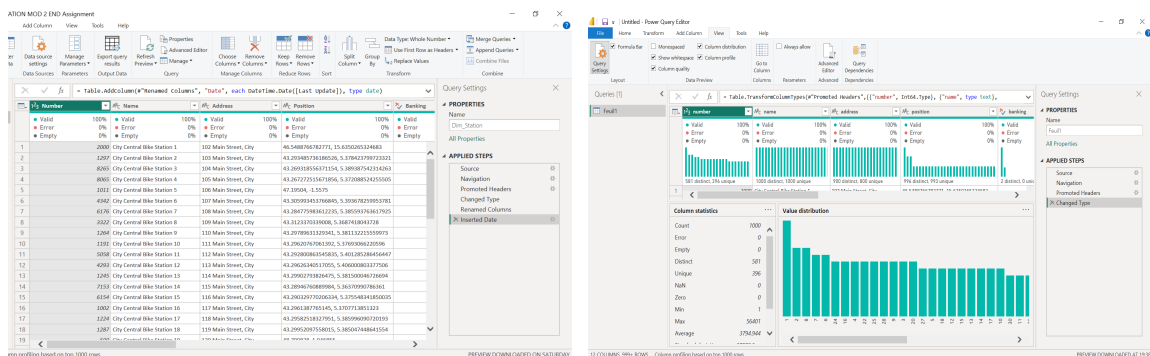
Public bike-sharing systems generate continuous data from hundreds of stations across different cities. The task is to analyze this real-time bike station dataset to understand station performance, usage efficiency, and operational patterns across cities.

Applying Power BI skills—from data cleaning and modeling to DAX and dashboarding—to turn raw data into actionable urban mobility insights.

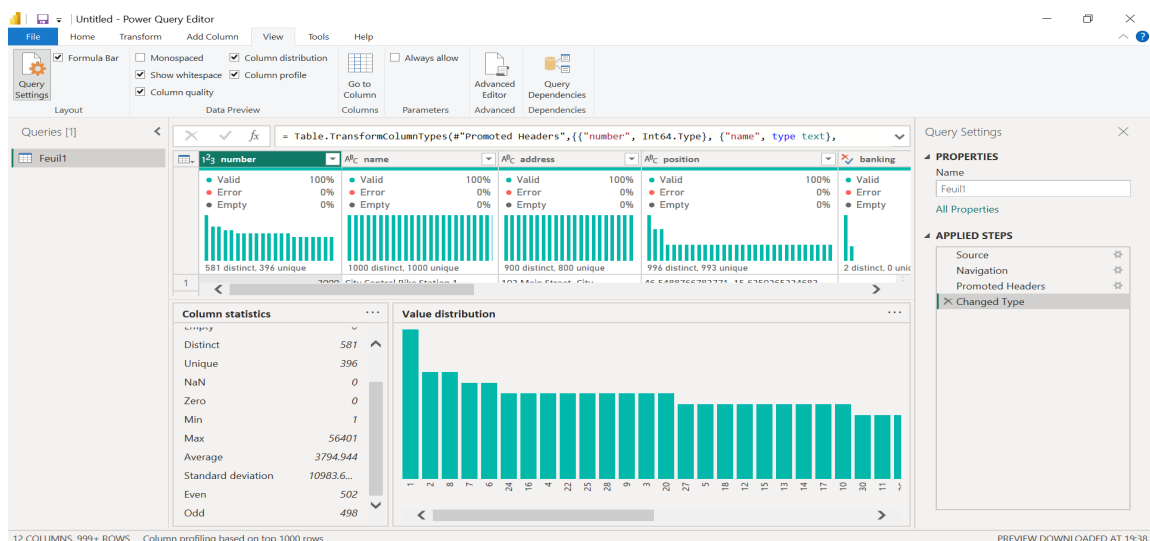
Dataset:

Source: [Dataset link](#)

- Import and explore the dataset in Power BI.



- Clean and transform data using Power Query.



POWER BI BIKE STATION MOD 2 END Assignment

Queries [1] Fact_BikeStation

Table: Fact_BikeStation

Ranking	City Name	City Address	City Position	City Type	City Status
1	1716	1716	1716	1716	1716
2	1716	1716	1716	1716	1716
3	1716	1716	1716	1716	1716
4	1716	1716	1716	1716	1716
5	1716	1716	1716	1716	1716
6	1716	1716	1716	1716	1716
7	1716	1716	1716	1716	1716
8	1716	1716	1716	1716	1716
9	1716	1716	1716	1716	1716
10	1716	1716	1716	1716	1716
11	1716	1716	1716	1716	1716
12	1716	1716	1716	1716	1716
13	1716	1716	1716	1716	1716
14	1716	1716	1716	1716	1716
15	1716	1716	1716	1716	1716
16	1716	1716	1716	1716	1716
17	1716	1716	1716	1716	1716
18	1716	1716	1716	1716	1716
19	1716	1716	1716	1716	1716
20	1716	1716	1716	1716	1716

POWER BI BIKE STATION MOD 2 END Assignment

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1	1716	1716	1716	1716	1716
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3	1716	1716	1716	1716	1716
4	1716	1716	1716	1716	1716
5	1716	1716	1716	1716	1716
6	1716	1716	1716	1716	1716
7	1716	1716	1716	1716	1716
8	1716	1716	1716	1716	1716
9	1716	1716	1716	1716	1716
10	1716	1716	1716	1716	1716
11	1716	1716	1716	1716	1716
12	1716	1716	1716	1716	1716
13	1716	1716	1716	1716	1716
14	1716	1716	1716	1716	1716
15	1716	1716	1716	1716	1716
16	1716	1716	1716	1716	1716
17	1716	1716	1716	1716	1716
18	1716	1716	1716	1716	1716
19	1716	1716	1716	1716	1716
20	1716	1716	1716	1716	1716

Build a logical data model.

Set up a fact table.

POWER BI BIKE STATION MOD 2 END Assignment

Queries [2] Fact_BikeStation

Table: Fact_BikeStation

Ranking	City Name	City Address	City Position	City Type	City Status
1	1716	1716	1716	1716	1716
2	1716	1716	1716	1716	1716
3	1716	1716	1716	1716	1716
4	1716	1716	1716	1716	1716
5	1716	1716	1716	1716	1716
6	1716	1716	1716	1716	1716
7	1716	1716	1716	1716	1716
8	1716	1716	1716	1716	1716
9	1716	1716	1716	1716	1716
10	1716	1716	1716	1716	1716
11	1716	1716	1716	1716	1716
12	1716	1716	1716	1716	1716
13	1716	1716	1716	1716	1716
14	1716	1716	1716	1716	1716
15	1716	1716	1716	1716	1716
16	1716	1716	1716	1716	1716
17	1716	1716	1716	1716	1716
18	1716	1716	1716	1716	1716
19	1716	1716	1716	1716	1716
20	1716	1716	1716	1716	1716

POWER BI BIKE STATION MOD 2 END Assignment

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2	1716	1716	1716	1716	1716
3	1716	1716	1716	1716	1716
4	1716	1716	1716	1716	1716
5	1716	1716	1716	1716	1716
6	1716	1716	1716	1716	1716
7	1716	1716	1716	1716	1716
8	1716	1716	1716	1716	1716
9	1716	1716	1716	1716	1716
10	1716	1716	1716	1716	1716
11	1716	1716	1716	1716	1716
12	1716	1716	1716	1716	1716
13	1716	1716	1716	1716	1716
14	1716	1716	1716	1716	1716
15	1716	1716	1716	1716	1716
16	1716	1716	1716	1716	1716
17	1716	1716	1716	1716	1716
18	1716	1716	1716	1716	1716
19	1716	1716	1716	1716	1716
20	1716	1716	1716	1716	1716

Remove Top Rows

Remove Bottom Rows

Remove Alternate Rows

Remove Duplicates

Remove Blank Rows

Remove Errors

Create dimension tables and

Create Dim_date

POWER BI BIKE STATION MOD 2 END Assignment

Queries [3] Dim_Date

Table: Dim_Date

Ranking	City Name	City Address	City Position	City Type	City Status
1	1716	1716	1716	1716	1716
2	1716	1716	1716	1716	1716
3	1716	1716	1716	1716	1716
4	1716	1716	1716	1716	1716
5	1716	1716	1716	1716	1716
6	1716	1716	1716	1716	1716
7	1716	1716	1716	1716	1716
8	1716	1716	1716	1716	1716
9	1716	1716	1716	1716	1716
10	1716	1716	1716	1716	1716
11	1716	1716	1716	1716	1716
12	1716	1716	1716	1716	1716
13	1716	1716	1716	1716	1716
14	1716	1716	1716	1716	1716
15	1716	1716	1716	1716	1716
16	1716	1716	1716	1716	1716
17	1716	1716	1716	1716	1716
18	1716	1716	1716	1716	1716
19	1716	1716	1716	1716	1716
20	1716	1716	1716	1716	1716

Create Dim_Station

POWER BI BIKE STATION MOD 2 END Assignment

File Home Transform Add Column View Tools Help

Close & Apply New Recent Enter Data source settings Manage Parameters Output Data Query Properties Advanced Editor Choose Remove Keep Remove Split Group Data Type Whole Number Merge Queries Append Queries Combine Files

Queries [2] Fact_BikeStation Dim_Station

table.Distinct(*Removed Columns*, {"Number"})

	Number	Name	Address	Position	Banking
	Valid	Valid	Valid	Valid	Valid
	Error	Error	Error	Error	Error
	Empty	Empty	Empty	Empty	Empty
1	2000	City Central Bike Station 1	102 Main Street, City	46.5488766782771, 12.620265324983	
2	1767	City Central Bike Station 2	103 Main Street, City	43.293485736186576, 5.37842379973331	
3	8765	City Central Bike Station 3	104 Main Street, City	43.269318556371154, 5.38038754214263	
4	8066	City Central Bike Station 4	105 Main Street, City	43.267272515671856, 5.37088248255005	
5	1011	City Central Bike Station 5	106 Main Street, City	47.32904, 1.5075	
6	4342	City Central Bike Station 6	107 Main Street, City	43.305483453768845, 5.39367825993781	
7	6176	City Central Bike Station 7	108 Main Street, City	43.284775982612235, 5.385593793617925	
8	3232	City Central Bike Station 8	109 Main Street, City	43.3133370339008, 5.3687418043728	
9	1764	City Central Bike Station 9	110 Main Street, City	43.29789631329341, 5.38113215559973	
10	1191	City Central Bike Station 10	111 Main Street, City	43.29620767061392, 5.37693066220596	
11	5050	City Central Bike Station 11	112 Main Street, City	43.292800863545835, 5.401282284546447	
12	4293	City Central Bike Station 12	113 Main Street, City	43.29626340517055, 5.406000803377506	
13	1245	City Central Bike Station 13	114 Main Street, City	43.29902793826475, 5.381500046728694	
14	7153	City Central Bike Station 14	115 Main Street, City	43.2894676889984, 5.3637090786361	
15	6154	City Central Bike Station 15	116 Main Street, City	43.290329770206334, 5.375548341850035	
16	1002	City Central Bike Station 16	117 Main Street, City	43.2961387760445, 5.3707713851123	
17	1224	City Central Bike Station 17	118 Main Street, City	43.29582518327951, 5.385996090720193	
18	1287	City Central Bike Station 18	119 Main Street, City	43.29952097558015, 5.385047448641554	
19	4001	City Central Bike Station 19	120 Main Street, City	43.29952097558015, 5.385047448641554	

8 COLUMNS, 999+ ROWS Column profiling based on top 1000 rows

PREVIEW DOWNLOADED ON SATURDAY

- Defining the relationships between the fact and dimension tables

Dim_Station[Number] 1 ————— * Fact_BikeStation[Number]

POWER BI BIKE STATION MOD 2 END Assignment • Last saved: Yesterday at 9:52 PM

File Home Help

Clipboard Get data Excel workbook OneLake catalog SQL Server Data Recent sources

Transform Refresh data Queries Relationships Manage relationships New measure column New table Calculations New parameter Parameters Manage roles View as Security Q&A Language setup Q&A Linguistic schema Sensitivity Public

Properties

General

Name Fact_BikeStation

Description Enter a description

Synonyms fact bike station, Fact_BikeStation, station, bike station

Row label Select a row label

Is hidden

Data

Tables

Fact_BikeStation

Address

Available Bike Stands

Available Bikes

Banking

Bike Stands

Bonus

Contract Name

Date

Last Update

Name

Number

Position

Status

All tables

Dim_Date[Date] 1 ————— * Fact_BikeStation[Date]

POWER BI BIKE STATION MOD 2 END Assignment • Last saved: 9/26/2026 at 9:52 PM

File Home Help

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Contract Name

Date

Last Update

Name

Number

Position

Status

Dim_Date

Date

Day Name

Day Update

Month Name

Quarter

Year

Dim_Station

Address

Banking

Bonus

Contract Name

Name

Number

Position

Status

All tables

Relationship between Fact and dimension table flow chart

Dim_Station



Fact_BikeStation



Dim_Date

- Create **DAX measures** and calculated columns (wherever needed).

Create DAX Measures

1. Total Bike Capacity

Total Bike Capacity =SUM(Fact_BikeStation[Bike stands])

2. Available Bikes

Total Available Bikes = SUM(Fact_BikeStation[Available bikes])

3. Available Bike Stands

Total Available Stands = SUM(Fact_BikeStation[Available Bike stands])

4. Used Bikes

Used Bikes =[Total Bike Capacity] - [Total Available Bikes]

5. Bike Usage Rate

Bike Usage Rate =DIVIDE([Used Bikes],[Total Bike Capacity],0)

After creating it: **Select the measure** → **Measure tools** → **Format** → **Percentage**
Set it to **1 or 2 decimal places**.

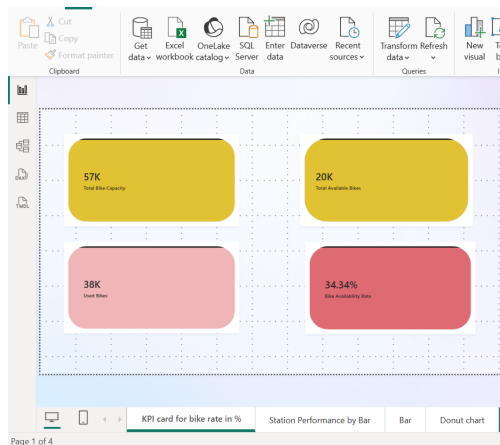
6. Bike Availability Rate

Bike Availability Rate = $\text{DIVIDE}([\text{Total Available Bikes}], [\text{Total Bike Capacity}], 0)$

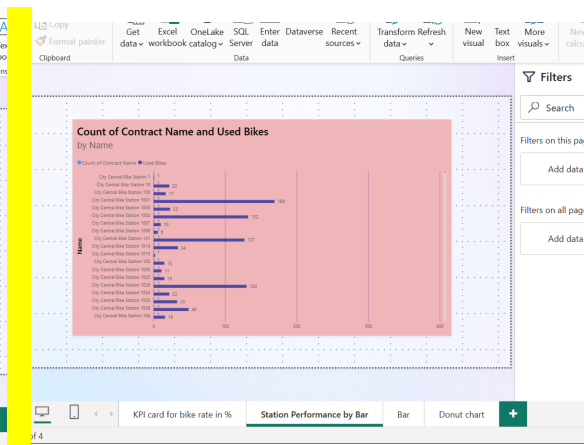
Set to **Percentage**

- **Design interactive visuals and reports/dashboards.**

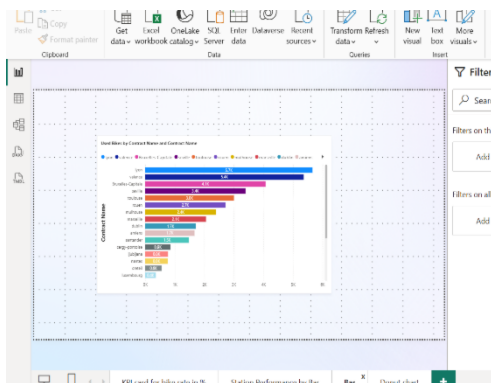
KPI CARD



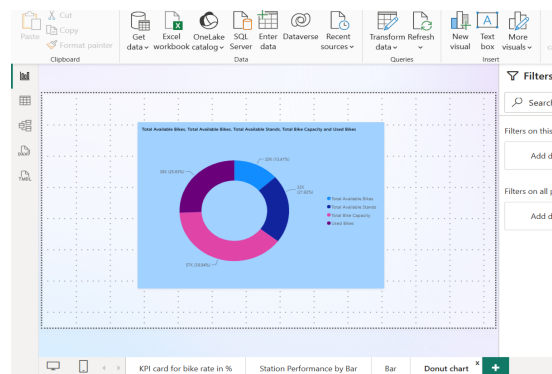
BAR CHART



CLUSTER BAR



DONUT CHART



Conclusion

The **Bike Station Sharing Power BI** project successfully analyzed bike station data to understand station utilization, bike availability, capacity, and operational status across different cities or contract areas.

The data was cleaned and transformed using **Power Query**, followed by the creation of a structured data model using **Fact_BikeStation**, **Dim_Station**, and **Dim_Date** tables. DAX measures were developed to calculate total bike capacity, available bikes, used bikes, bike usage rate, and availability rate.

The interactive Power BI dashboard provides a clear view of **city-wise usage, station performance, available versus used bikes, usage trends, and station status**. Interactive slicers allow users to filter the analysis by city, station, date, and status, making the dashboard useful for exploring different operational scenarios.

Overall, this project demonstrates how **Power BI, Power Query, data modeling, DAX, and interactive visualizations** can transform raw bike-sharing data into meaningful insights. The analysis can support better **station monitoring, capacity planning, bike redistribution, and operational decision-making**.